Rover power options

Race tracker boat unit · 14 Oct 2026 · prepared for the fabricator call

1. Scope and assumptions

- **Boat:** Pi Zero 2 W plus the ArduSimple simpleRTK2B (ZED-F9P, XBee in its socket). A 3A+ is shown once as a reference.
- **Day:** unit off 14 h overnight, then 10 h on: 4 races of 40 min (2.67 h racing) and 7.33 h on but not racing.
- **Packs (usable, about 80% of nominal at 3.7 V):** 10,000 mAh bank about 30 Wh; 20,000 mAh about 59 Wh; 26,800 mAh about 79 Wh. “Days” means days of this schedule between charges.
- **What is measured** (FNIRSI FNB58, 13 Oct 2026): the 3A+ boat draws **1.73 W**; the Pi 3A+ alone **0.94 W**; so the ArduSimple side is about **0.79 W**. The Pi Zero 2 W alone was also measured at **0.6 W** (state to be confirmed: agent running or idle, Wi-Fi on or off). The Zero 2 W boat at about 1.4 W is that 0.6 W plus the measured 0.79 W. **Everything else is an estimate:** a halted Pi at 0.1 W, the off-state draws, and the trims.
- **Costs** are per rover in GBP from the fabricators’ preliminary BOM (11 Oct 2026, ex VAT, planning rates for about 100 units) where one exists; otherwise “not quantified”. Design NRE is a shared cost: £43,000–49,000, plus about £7,000 prototype materials and about £20,000 for US lab work if approval is needed.

Their report budgets the rover at 3.1–3.5 W (Pi 1.75 W). The measured 3A+ boat is 1.73 W. The report states its budget is not a measured roll-up. The Zero 2 W Pi figure is now measured (0.6 W), but a whole Zero 2 W boat has not been measured end to end.

2. All options compared

Option	Saves racing	Saves between races	Wh/day	10,000 mAh	20,000 mAh	26,800 mAh
Baseline: Zero 2 W, off overnight, Pi and Ardu on between races	–	–	14.3	2.1	4.1	5.5
1. Radio sleep between races (XBee and ZED asleep, Pi on)	0	0.30 W	12.1	2.5	4.9	6.5
2. Trim the Pi (Wi-Fi, HDMI, LEDs off, fewer cores)	0.15 W	0.15 W	12.8	2.3	4.6	6.2
3. Halt the Pi between races, Ardu left powered	0	0.51 W	10.5	2.8	5.6	7.5
4. Everything off between races, woken by a supervisor	0	1.37 W	4.2	7.1	13.9	18.7
5. Own ZED board without the linear regulator	0.20 W	0.20 W	12.3	2.4	4.8	6.4
6. Second radio (a cost)	–0.10 W	–0.10 W	15.3	2.0	3.9	5.2
7. Minimum Pi setup: trims plus radio sleep	0.15 W	0.45 W	10.6	2.8	5.6	7.5

8. Stack: trims, radio sleep and own board	0.35 W	0.65 W	8.6	3.5	6.9	9.2
Reference: 3A+ instead of Zero 2 W (baseline strategy)	–0.33 W (costs more)	–0.33 W (costs more)	17.6	1.7	3.4	4.5

Days are days per charge on the Zero 2 W boat, except the 3A+ reference row. All rows also draw about 0.02 W overnight with the unit off. Savings are against the baseline row. Rows 7 and 8 combine levers, so their savings are not the sum of the single rows where they overlap.

3. Each option

1. Radio sleep between races

What it does	Base sends a power frame; the rover powers its XBee down (SLEEP_RQ pin) and the ZED into software backup (UBX-RXM-PMREQ). The Pi stays on and keeps logging. A base wake frame, repeated for one listen cycle, ends it.
Power saved	About 0.30 W between races (XBee 0.09 W plus ZED 0.2 W, datasheet figures). Nothing while racing.
Days per charge	2.5–6.5 days (2.5, 4.9, 6.5) against 2.1–5.5 for the baseline.
Cost	About £0: one wire (GPIO17 to XBee pin 9) and a 10 kΩ pull-down. Software is built.
Benefit	Cheap, remote start and stop from the base dashboard, frees the boat's transmit slot while asleep.
Risks and open items	Whether XBee pin 9 is reachable on the ArduSimple socket is unconfirmed. The ZED refuses backup while its USB port is connected unless `GPS_SLEEP_FORCE_USB=1`. Each wake costs a GPS reacquisition; a hot start needs `V_BCKP` kept powered.

2. Trim the Pi

What it does	Wi-Fi off during racing, HDMI and LEDs off, fewer or slower cores.
Power saved	About 0.15 W, range 0.05–0.3 W. A guess built from figures for older Pis; none measured on a Zero 2 W. Applies while racing as well.
Days per charge	2.3–6.2 days (2.3, 4.6, 6.2).
Cost	£0.
Benefit	The only option here that helps during a race besides the Zero 2 W swap and the board change (5).
Risks and open items	With Wi-Fi off the boat's dashboard and uploads are unreachable until it is switched back on. The saving is the least certain in this document.

3. Halt the Pi between races, Ardu left powered

What it does	A new halt action on the existing power frame (about an hour of code) runs the same `shutdown` routine the scheduled shutdown uses. The Ardu stays powered through the Pi's USB port, so the GPS keeps its fix.
Power saved	About 0.5 W between races (Pi 0.6 W less an assumed 0.1 W halted draw, unmeasured).
Days per charge	2.8–7.5 days (2.8, 5.6, 7.5).
Cost	£0 in hardware, but it needs a wake source: a button, an RTC alarm board, or GPIO3 pulled low. A halted Pi cannot be woken by radio.
Benefit	Fast race start (20–30 s boot, GPS already locked).
Risks and open items	A stray or corrupted frame must not halt boats mid-race: require explicit boat ids, a confirm, and refuse unless stationary. Logging stops while halted.

4. Everything off between races, woken by a supervisor

What it does	An always-on microcontroller reads the radio's UART in cyclic sleep, recognises the base's wake frame, and enables the Pi and GNSS rails. This is the fabricators' supervisor-and-load-switch architecture.
Power saved	About 1.37 W between races: the unit draws about 0.03 W (supervisor about 5 mA, radio in cyclic sleep, ZED backup 1.4 mA).
Days per charge	7.1–18.7 days (7.1, 13.9, 18.7). The only option that changes runtime by a large factor.
Cost	In their BOM per rover: supervisor MCU £5.00, five load switches £2.10, fuel gauge £2.80, charger £2.20, regulators about £7. Their whole motherboard totals £450 (one radio) to £510 (two radios). The wake-frame firmware is not in their scope and is ours.
Benefit	Remote start of boats from the base, near-zero standby, and the architecture the report already proposes.
Risks and open items	The GNSS cold-starts before each race unless V_BCKP holds its data, and an RTK fix can take minutes. The supervisor must be able to read the radio's UART while the Pi is off. The ZED-side saving needs the ArduSimple powered separately from the Pi.

5. Own ZED board without the linear regulator

What it does	Replace the ArduSimple with a board that powers the ZED, antenna and radio from a switch-mode rail instead of its 5 V to 3.3 V linear path (66% efficient, per the report).
Power saved	About 0.2 W (range 0.17–0.27 W), racing and between races. By their own loads (ZED 0.297 W, antenna 0.165 W, radio 0.096 W) the linear path wastes about 0.22 W.
Days per charge	2.4–6.4 days (2.4, 4.8, 6.4).
Cost	ArduSimple board is £140 per rover in their BOM. A custom replacement is not quantified; the report advises against it at this volume (development risk unlikely to repay its saving). A cheaper variant is to bypass only the linear path, if the ArduSimple allows.
Benefit	Always on, so it also helps during a race.
Risks and open items	GNSS RF layout, certification, schedule. Whether a carrier can feed the ZED's 3.3 V directly is unanswered.

6. Second radio (a cost, not a saving)

What it does	A dedicated receive-only radio for corrections beside the telemetry radio, so telemetry is not blocked during the correction burst.
Power	About –0.10 W (29 mA at 3.3 V, datasheet), racing and between races.
Days per charge	2.0–5.2 days (2.0, 3.9, 5.2), about 0.1–0.4 days lower than the baseline.
Cost	£48 per radio (XBP9B S3B); the report totals £510 per rover with two radios against £450 with one (about £60 more), plus a second antenna.
Benefit	Capacity gain is small (about 80 ms of a 2 s cycle, 4%). The main benefit is correction rate decoupled from telemetry, and simpler timing.
Risks and open items	Two 900 MHz radios centimetres apart can desensitise each other; this needs an isolation estimate (channel masks, antenna spacing, filtering) before it can be compared with one radio.

7 and 8. Combinations

7. Minimum Pi setup	Trims plus radio sleep between races: about 0.15 W racing and about 0.45 W between races saved. 2.8–7.5 days (2.8, 5.6, 7.5). This lands close to halting the Pi (option 3) without needing any wake mechanism, but rests on the unmeasured trims.
8. Stack	Trims, radio sleep and an own board that removes the linear loss: about 0.35 W racing and about 0.65 W between races. 3.5–9.2 days (3.5, 6.9, 9.2).
Reference: 3A+	Staying on a 3A+ costs about 0.33 W more than a Zero 2 W in every state, giving 1.7–4.5 days (1.7, 3.4, 4.5) on the baseline strategy.

4. Observations

- Only option 4, switching the whole unit off between races, changes runtime by a large factor, because it is the only one that removes the Ardu's 0.79 W.
- The sleep and halt options (1–3) only help between races. The boat must be awake to race.
- During a race, only the Pi trims, the Zero 2 W swap and the board change reduce draw, and each is small or unquantified.
- With nightly charging, any pack here covers a day easily (about 4–15 Wh a day), so larger packs matter only when charging between days is not possible.

5. What to measure next

1. A whole Zero 2 W boat on the meter, running the real agent (confirms the 0.6 W Pi and the combined 1.4 W).
2. The Pi halted, with and without the Ardu (the 0.1 W guess).
3. Wi-Fi off, HDMI off and LEDs off as separate runs (the 0.15 W trims).
4. The XBee asleep and the ZED in backup (the 0.3 W radio sleep).
5. A run-down test on the real pack, which includes the power bank's light-load cutoff behaviour.

Sources: fabricator report RPT21057-V01 (11–12 Oct 2026), FNIRSI FNB58 logs run1–run3 (13 Oct 2026), Digi datasheet figures as quoted in the project README, and published Pi Zero 2 W measurements (Jeff Geerling, CNX Software, Tom's Hardware), now superseded for the Pi by a direct measurement. The README identifies the XBee in the ArduSimple socket as an XBee SX, while the report's BOM lists XBP9B S3B (900HP); datasheet figures should be re-checked against the actual module.